higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONSTRUCTION PLANNING
NQF LEVEL 2

27 FEBRUARY 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1
1.1.1 Graphometer
1.1.2 Deviation
1.1.3 Curing
1.1.4 Ratio
1.1.5 Renovation

(5 × 1) (5)

1.2
- To ensure the safety and health of the occupants
- To ensure the uniformity in the law relating to the erection of buildings (2)

[7]

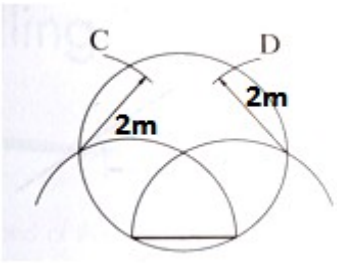
QUESTION 2

2.1
2.1.1 True
2.1.2 False
2.1.3 True
2.1.4 False
2.1.5 False

(5 × 1) (5)

2.2 Level the instrument at one point.✓
Keep the leveling staff at the other point where the distance need to be found.✓
Take the top and bottom stadia hair readings.✓
The distance = (top hair reading –bottom hair reading) × 100.✓ (4)

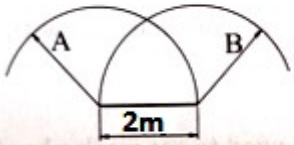
2.3

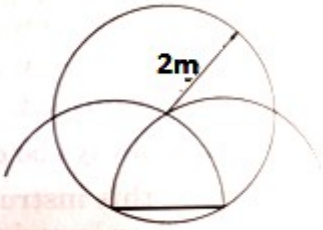




Draw a line 2 m long
Draw 2 arcs A,B, from both ends two meter long✓

from the point the arc meets draw a circle of 2 m radius✓





Draw an arc of 2 m to cut the Circle, from where Arc A and B cuts the circle✓

join all the points to get the hexagon✓

(6)

2.4
- Kinks in the pipe can cause incorrect level
- Air leaks can cause faulty readings
- The water can leak and can cause incorrect reading
- The water can spill if one end is lifted too high
- Surface tension can cause incorrect reading (Any 4 × 1) (4)

2.5 Elevation of the above bench mark = 100 + 1,3✓= 101,3 m✓
Elevation of the point B = 101,3 – 0,95✓ = 100,35 m✓ (4)

2.6 1 – Ext measuring jaws
2 – Internal measuring jaws
3 – Depth gauge/rod
4 – Locking screw
5 – Main scale (6)
6 – Vernier scale (6 × 1) **[29]**

QUESTION 3

3.1
- To determine the capacity of the ground to support
- To determine what kind of foundation is best for the structure
- To determine if there is any contamination (3)

3.2
- For low bearing capacity ground
- For multi stories buildings
- Where the columns are going to occupy most of the site
- When building on poor soil (4)

3.3
- To distribute the load evenly to the ground
- To avoid any settlement or movement of the building
- To provide a level surface to start work
- It anchors the building to the ground (Any 3 × 1) (3)

[10]

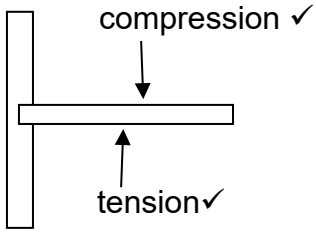
QUESTION 4

4.1
4.1.1 Key stone
4.1.2 Torque
4.1.3 Shear force
4.1.4 Cantilever
4.1.5 Short column
(5 × 1) (5)

4.2 4.2.1 A column with a small cross sectional area compared to its length will bend (bulk) due to excess load. (2)

4.2.2 Bracing or supporting the column at the centre. (1)

4.3



(2)

4.4 AC = strut
BC = tie
CD = strut
DA = tie (4 × ½) (2)

4.5 •If the load is concentrated at one point on the beam it is called Point load✓
•If the load is spread over the entire beam or over certain section it is called uniform distributed load✓ (2)

4.6 Plane frames – are pin jointed frames in only 2 dimensions✓
Space frames – are pin jointed frames in 3 dimensions✓ (2)

4.7 Five hinge failure – when the point load acts at the center✓
Four hinge failure – when the point load acts on one side of the arch✓ (2)
[18]

QUESTION 5

5.1 5.1.1 7×10^{-10}
5.1.2 160000000 millimeters
5.1.3 8 700 seconds
5.1.4 $2{,}5 \times 10^{-8}$
5.1.5 81º 14’
(5 × 1) (5)

5.2 5.2.1 Newton or N
5.2.2 Bar or Pa
5.2.3 Ampère or A
5.2.4 Joule or J
5.2.5 Watt or W
(5 × 1) (5)
[10]

QUESTION 6

6.1 Perimeter of the wall = (8 × 2) + (4,5 × 2) = 25 m✓
Volume of concrete = 25 × 0,6 × 0,2✓
= 3 m²✓ (3)

6.2 Area of the wall = 25 × 3,2✓
= 80 m² ✓

Opening of 4 windows = 4 × 2 × 2 = 16 m²✓
Opening of door = 2 × 2,8 = 5,6 m²✓

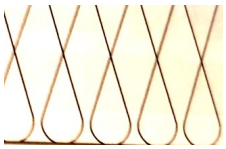
Net area for bricklaying = 80 – (16+ 5,6) = 58,4 m²✓ (5)

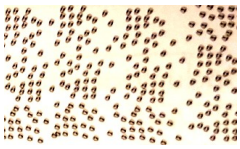
6.3 Area for painting = 58,4 × 2 = 116,8 m²✓
Paint needed = 116,8/4 = 29,2 = 30 L✓ (2)
[10]

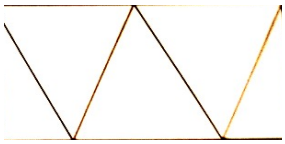
QUESTION 7

7.1


STONE


INSULATION


SCREED


HARD CORE


BRICK (5)

7.2 7.2.1 Rain water pipe
7.2.2 Built in cupboard
7.2.3 Soil pipe
7.2.4 Half round
7.2.5 Storm water pipe
7.2.6 Finished floor level

(6 × 1) (6)

7.3 7.3.1 Drawings, diagrams and written notes that show the construction team what to build

7.3.2 Drawings prepared by an engineer showing reinforced structural members

7.3.3 Drawings that give information on services to the building, including sewer, plumbing electrical etc.

(3 × 1) (3)

7.4
- Site plan
- Lay out plan

(2)
[10]

TOTAL: 100

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